Vinay Kumar

Gaya, Bihar | Haldia, West Bengal
+91-8830603531 rajupendra15@gmail.com LinkedIn GitHub LeetCode Portfolio

Education

Haldia Institute of Technology 2024 – 2028
B.Tech in Computer Science and Engineering — CGPA: 8.46 *Haldia, West Bengal*

A.M College Gaya 2023
12th, BSEB — Percentage: 83% *Gaya, Bihar*

High School Rasalpur Manpur Gaya 2021
10th, BSEB — Percentage: 81% *Gaya, Bihar*

Technical Skills

Languages: Java, Python, JavaScript, SQL, HTML5, CSS3
Frameworks & Technologies: React.js, Next.js, Node.js, Express.js, FastAPI, REST APIs, WebSockets
Databases & Cloud: MongoDB, Redis, Firebase (Authentication, Cloud Firestore, Storage)
AI & Computer Vision: Generative AI, RAG, OpenAI API, Prompt Engineering, Machine Learning, OpenCV
Core: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks
Tools: Git, GitHub, VS Code, Postman, Vite, Vitest

Internships

Full Stack Developer Intern Apr 2026 – Present
HSR Hi-Tech Solutions
- Contributed to full-stack web applications using React.js, Node.js, Express.js, MongoDB, and Firebase, building responsive user interfaces and implementing application functionality.
- Integrated backend services and participated in real-world projects and technical discussions.

Projects

CrowdOS – AI-Powered Crowd Intelligence Platform GitHub
- Developed an AI-powered platform to monitor crowd movement, track occupancy, and identify potential crowd-risk situations from CCTV/IP camera feeds.
- Designed a modular architecture with Next.js, FastAPI, MongoDB, Redis messaging, and an independent AI engine.
- Integrated OpenCV, YOLO, and ByteTrack; implemented entry/exit events, occupancy tracking, and live monitoring.
- Contributed to backend API integration, event-processing workflows, and predictive crowd-risk analysis.

WiFi Service Desk – SaaS Complaint Management System GitHub
- Built a SaaS complaint management platform for ISPs with customer, technician, and admin role-based interfaces.
- Built with React.js, Vite, Tailwind CSS, and Firebase (Authentication, Cloud Firestore, Storage), featuring real-time complaint tracking, technician and admin dashboards, and analytics using Recharts.
- Enforced role-based access with Firestore security rules for permissions and status transitions; tested with Vitest.

AI Learning OS – AI-Powered Learning Platform GitHub
- Developed AiAnalyzer, a platform that transforms YouTube educational videos into structured study materials using transcript extraction and AI-powered content analysis.
- Built a RAG-based AI tutor for contextual question answering, with automated generation of notes, flashcards, quizzes, learning roadmaps, and exam-focused revision summaries.
- Implemented background processing with Redis and Bull Queue, JWT authentication, personalized learning history, and Razorpay-based credit billing using Next.js, Node.js, Express.js, MongoDB, and the OpenAI API.

Positions of Responsibility

Full Stack & Machine Learning Mentor Present
HSR Hi-Tech Solutions
- Mentoring students in Full Stack Web Development and Machine Learning (React.js, Node.js, MongoDB, Python).
- Reviewing assignments, providing technical feedback, and conducting onboarding sessions for trainees.

Languages

English — Professional Working Proficiency | Hindi — Native